

Initial brief from your company head of product development:

The company your team works for (Trevelyan Interactive) has decided to produce an electronic version of a game based on the classic 1980s board game “Scotland Yard”, but with a twist.

Rather than Scotland Yard detectives hunting a fugitive through the streets of London, the detectives in your game are students searching for your lecturer around the mean streets of Leeds – the evil Dr. X has got your assessment feedback, and you are determined to hunt them down.

The product will have 3 components:

1. A game server
2. A mobile phone client app
3. A web app

Multiple teams are working on various parts of the product, and some co-ordination may be required.

The minimum viable product for your team is a mobile client app, allowing player to start, join and end games, to play as either the fugitive or the detective, and to make moves within the game.

It is up to you to decide how your app should behave – how much information should be available to the player and how it is displayed, and whether (and how) the app should assist the player.

You are free to use whatever technology you think is appropriate, however, the business will derive more value from your team up-skilling in certain technologies.

The **web app** is intended to supplement the client app, and be viewed on a larger screen to provide an overview of the entire map, and the positions of the students/detectives and (when revealed) the fugitive.

The **Server** will provide a RESTful API allowing game sessions to be initialised. After that, it will maintain games until they are complete, allowing clients to join games and play according to the rules.

When creating a game, it will ideally be possible to specify the number of rounds, and which map the game will use.

When you ask the server to close the lobby and start the game, it will allocate random starting positions on the map (from a pre-determined list) for both Dr. X (fugitive) and all the students (detectives), and assign the requisite number of tickets to the students.

The server *should* only allow you to make moves according to the rules of the game. See separate game overview document for the playing rules.

The server will keep track of whose turn it is, how many tickets players have, and whether Dr X's position should be revealed, and also keep a log of Dr X's use of tickets.

At any time a player (via the client app) should be able to ask for the state of the game and that player's current position.

To play the game, each player (via the client app) will request that the server make a particular move. The server will respond by allowing the move and recording that that player has taken their turn, or else disallow the move. It is not possible to 'pass' on a turn.

